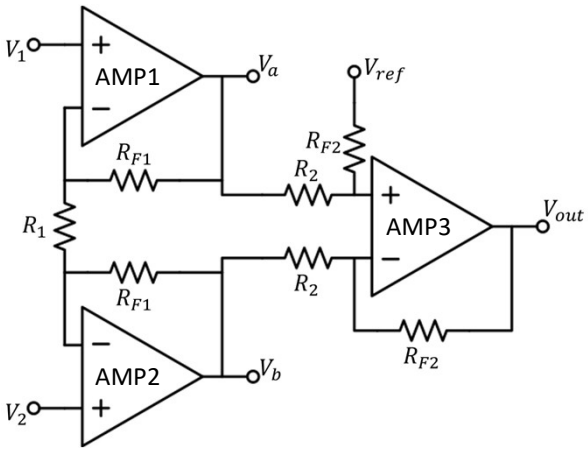
Exercice 7 / Aufgabe 7



Calculez V_a et V_b. Calculez V_{out} avec des expressions pour V_a et V_b.

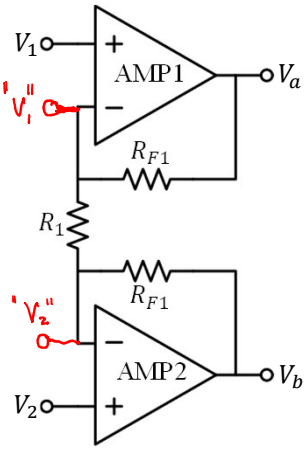
Berechnen Sie V_a und V_b. Berechnen Sie V_{out} mit Ausdrücken für V_a und V_b.

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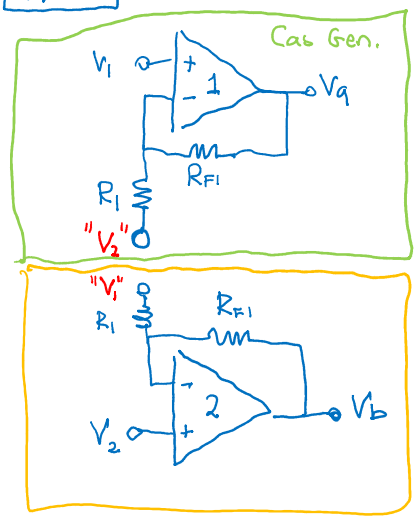
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Exercice 7 / Aufgabe 7

SOLUTION



$V_+ = V_-$



$$V_a = V_1 \frac{R_{F1}+R_1}{R_1} - V_2 \frac{R_{F1}}{R_1}$$

$$V_b = V_2 \frac{R_{F1}+R_1}{R_1} - V_1 \frac{R_{F1}}{R_1}$$

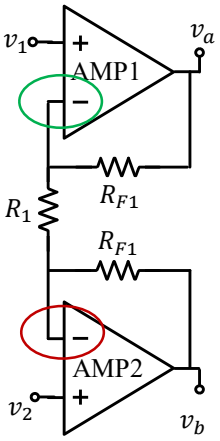
SOLUTION

SOL

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Exercice 7 / Aufgabe 7

SOLUTION

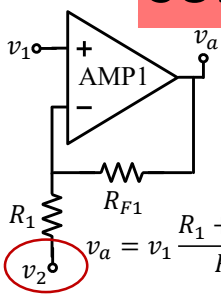


AMP1 *force* le nœud V_{-1} à la valeur de $V_{+1} = v_1$. Ce nœud peut être traité comme un nœud *basse-impédance*

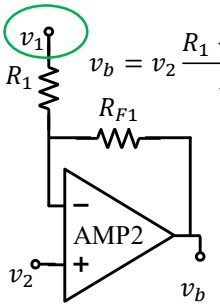
AMP1 zwingt den Knoten V_{-1} auf den Wert von $V_{+1} = v_1$. Dieser Knoten kann als niederohmiger Knoten behandelt werden

AMP2 *force* le noeud V_{-2} à la valeur de $V_{+2} = v_2$. Ce nœud peut être traité comme un nœud *basse-impédance*

AMP2 zwingt den Knoten V_{-2} auf den Wert von $V_{+2} = v_2$. Dieser Knoten kann als niederohmiger Knoten behandelt werden



$$v_a = v_1 \frac{R_1 + R_{F1}}{R_1} - v_2 \frac{R_{F1}}{R_1}$$



$$v_b = v_2 \frac{R_1 + R_{F1}}{R_1} - v_1 \frac{R_{F1}}{R_1}$$

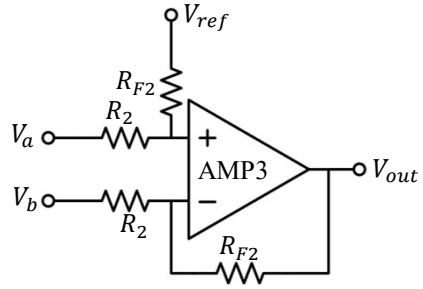
SOLUTION

SOL

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Exercice 7 / Aufgabe 7

SOLUTION



$$V_a = \frac{R_1 + R_{F1}}{R_1} V_1 - \frac{R_{F1}}{R_1} V_2 \qquad V_b = \frac{R_1 + R_{F1}}{R_1} V_2 - \frac{R_{F1}}{R_1} V_1$$

$$V_+ = V_a \frac{R_{F2}}{R_{F2} + R_2} + V_{ref} \frac{R_2}{R_{F2} + R_2}$$

$$V_{out} = V_a \frac{R_{F2}}{R_2} - V_b \frac{R_{F2}}{R_2} + V_{ref}$$

Cas général
Allgemeiner Fall

$$V_{out} = \frac{R_{F2}}{R_2}\left(\frac{R_1 + R_{F1}}{R_1} V_1 - \frac{R_{F1}}{R_1} V_2 - \frac{R_1 + R_{F1}}{R_1} V_2 + \frac{R_{F1}}{R_1} V_1\right)$$

$$V_{out} = V_+ \frac{R_{F2} + R_2}{R_2} - V_b \frac{R_{F2}}{R_2}$$

$$V_{out} = \frac{R_{F2}}{R_2} \frac{R_1 + 2R_{F1}}{R_1} (V_1 - V_2)$$

Ampli Différentiel avec très grand impédance d'entrée et une grande réjection de Mode Commun

Differenzverstärker mit sehr hoher Eingangsimpedanz und grosser Gleichtaktunterdrückung

SOLUTION

SOL

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SOLUTION

Amplificateur d'instrumentation: Ampli Différentiel avec très grand impédance d'entrée et une *grande réjection de Mode Commun*

Instrumentierungsverstärker: Differenzverstärker mit sehr hoher Eingangsimpedanz und *grosser Gleichtaktunterdrückung*

$$V_a = \frac{R_1 + R_{F1}}{R_1} V_1 - \frac{R_{F1}}{R_1} V_2 \qquad V_b = \frac{R_1 + R_{F1}}{R_1} V_2 - \frac{R_{F1}}{R_1} V_1$$

$$V_{CM} = \frac{V_1 + V_2}{2} \qquad V_d = V_1 - V_2$$

$$V_1 = V_{CM} + \frac{V_d}{2} \qquad V_2 = V_{CM} - \frac{V_d}{2}$$

$$V_a = \frac{R_1 + R_{F1}}{R_1}\left(V_{CM} + \frac{V_d}{2}\right) - \frac{R_{F1}}{R_1}\left(V_{CM} - \frac{V_d}{2}\right)$$

$$V_a = V_{CM} + \frac{R_1 + 2R_{F1}}{2R_1} V_d \qquad V_b = V_{CM} - \frac{R_1 + 2R_{F1}}{2R_1} V_d$$

$$V_a = \frac{V_1 + V_2}{2} + \frac{R_1 + 2R_{F1}}{2R_1}(V_1 - V_2) \qquad V_b = \frac{V_1 + V_2}{2} - \frac{R_1 + 2R_{F1}}{2R_1}(V_1 - V_2)$$

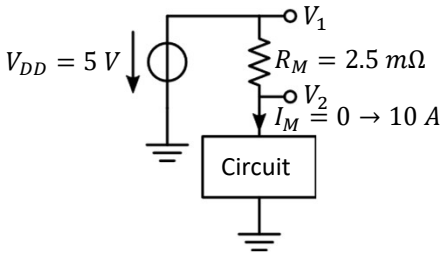
SOLUTION

SOL

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Example

SOLUTION



$$V_{out} = \frac{R_{F2}}{R_2}\frac{R_1 + 2R_{F1}}{R_1}(V_1 - V_2)$$

$$V_{out} = G_2 \times G_1(V_1 - V_2)$$

$V_M = 25\ mV \rightarrow 5V$

$G = 200 = G_1 \times G_2 = 14 \times 14$

$P_{R_M} = 10\ A \times 25\ mV = 250\ mW$

$$V_a = \frac{V_1 + V_2}{2} + \frac{R_1 + 2R_{F1}}{2R_1}(V_1 - V_2)$$

$$V_a = \frac{V_1 + V_2}{2} + G_1(V_1 - V_2)$$

$$\frac{V_1 + V_2}{2} = \frac{5\ V + (5\ V - 25\ mV)}{2} = 5\ V - 12.5\ mV$$

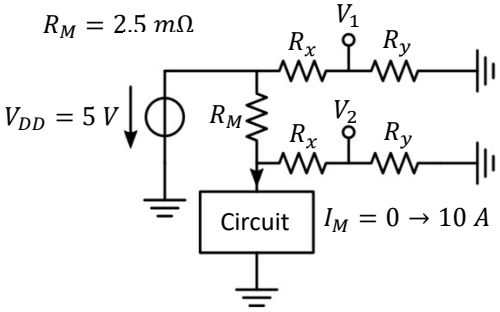
$$V_a = 5\ V - 12.5\ mV + 14(25\ mV) = 5.34\ V$$

SOLUTION

SOL

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SOLUTION



$$\frac{V_1 + V_2}{2} \approx 5\ V \times \frac{R_y}{R_y + R_x} = 4\ V$$

$$\frac{R_y}{R_y + R_x} = \frac{4}{5} = \frac{4}{4+1} = \frac{39\ k\Omega}{39\ k\Omega + 10\ k\Omega}$$

$V_M = 25\ mV \rightarrow 5V$

$$G = 200 \times \frac{39\ k\Omega + 10\ k\Omega}{39\ k\Omega} = 251 = G_1 \times G_2 = 15.8 \times 15.9$$

$$V_a = \frac{V_1 + V_2}{2} + G_1(V_1 - V_2)$$

$$\frac{V_1 + V_2}{2} = \frac{5\ V + (5\ V - 25\ mV)}{2} \times \frac{39\ k\Omega}{39\ k\Omega + 10\ k\Omega} = 3.97\ V$$

$$V_a = 3.97\ V + 15.8 \times (25\ mV) \times \frac{39\ k\Omega}{39\ k\Omega + 10\ k\Omega} = 4.28\ V$$

SOLUTION

SOL

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